

RAC PERFORMANCE DATA

Model Name : MSY-KY09VF-TH1 / MUJ-KY09VF-TH1

Power supply : 1 Ø 50 Hz 220 V

Comp.model : KVB053FXQMT

Ref.weight : (R32) = 350 g.

C / T size : C/T: Ø3.0-1.2-900L mm.

LEV :

Reference design specification no. : 702-RA-0116-19 / MCP-22B003-RFD-F

Indoor unit : MSY-KY09VF-TH1

-Production date : 12-Mar-25 Line : A

-Lot size : 5011296-5011695 Units 400

Outdoor unit : MUJ-KY09VF-TH1

-Production date : 8-Mar-25 Line : C

-Lot size : 5011340-5011639 Units 300

Period of test : 18-27 Mar '25

APPROVE (QA Dept. Mgr.)	APPROVE (Dev. Test SM.)	CHECKER	RECORD
REPORT NO.	SAM-25-760-0015		

Reference Product standard : ISO 5151 : 1994 , JIS-C9612 , TIS 2134-2553

Reference Safety test standard : TIS 1155-2557 , TIS 1529-2561

Reference Capacity and Performance test standard : ISO 5151 : 1994 , JIS-C9612 , TIS 2134-2553

Reference EMC test standard : TIS 1155-2557 , TIS 1529-2561

Reference Sound test standard : ISO 3744:2010 , ISO 3745:2010

Reference Other test standard :

1. Test result :

Classification of Evaluate Item		Specification and Tolerance		Q'ty	Judg	Data test result	
Test	Item	Condition					
Cooling Capacity and General Performance	Cooling capacity [Rate]	Room temp. In DB27.0 °C, WB19.0 °C Out DB35.0 °C, WB24.0 °C	Spec. 1 φ , 50 Hz, 220 V Capacity= 2700 W Power input= 865 W Current= 4.50 A EER= 3.12 W/W SEER (Label)= 18.15 Btu/h.w EGAT Energy rating = No.5 Pipe length = 7.5 m.	Tolerance ≥ 97.2 % ≤ 110 % ≤ - % ≥ - % ≥ 95.2 % and ≥ 17.06 SEER ≥ 17.06	1	OK	S/N : 50111313 / 5011639 [RA-22] ✓ Capacity= 2835.8 W (105.030%) ✓ Power input= 872.3 W (100.844%) ✓ Current= 4.45 A (98.889%) ✓ EER= 3.251 W/W (104.199%) ✓ SEER (Label)= 19.52 Btu/h.w (107.548%) ✓ EGAT Energy rating = 5 ★ ✓ Pipe length = 7.5 m. ✓
	Cooling capacity [Half load]	Room temp. In DB20.0 °C, WB- °C Out DB7.0 °C, WB6.0 °C	Spec. 1 φ , 50 Hz, 220 V Capacity= 1417.9 W (Rate / 2) Power input= 284 W EER= 4.75 W/W Pipe length = 7.5 m.	Tolerance Q ± 141.8 W and ± 5 % of actual rate ≤ - % ≥ - % m.	1	OK	S/N : 50111313 / 5011639 [RA-22] ✓ Capacity= 1422.4 W (100.159%) ✓ Power input= 277.0 W (97.535%) ✓ EER= 5.135 W/W (108.105%) ✓ Pipe length = 7.5 m. ✓
	Cooling Overload	In 32.0 / 23.0, Out 43.0/26.0 °C	During 2 Hr. protector does not work and Pd< 4.17 MPa.		-	OK	Reference : MSY-KY09VF-TH1 (SAM-24-760-0028) ✓
	Starting test	In 32.0 / 23.0, Out 43.0/26.0 °C	Low voltage start ≤ 176 V		-	OK	Reference : MSY-KY09VF-TH1 (SAM-24-760-0028) ✓
	Freezing disturbance	In 21.0 / 15.0, Out 21.0/15 °C	Ice on indoor coil surface not over 10% of indoor coil area		-	OK	Reference : MSY-KY09VF-TH1 (SAM-24-760-0028) ✓
	Freezing dropping	In 21.0 / 15.0, Out 21.0/15 °C	6Hr. No ice & water drop from air outlet		-	OK	Reference : MSY-KY09VF-TH1 (SAM-24-760-0028) ✓
	Heating overload	In 27.0 / - , Out 24.0 / 18.0 °C	Voltage ± 10% operate 2 Hr. protector does not work				
	Automatic defrost	Low temp High Humidity ratio In 20.0 / 12.0, Out 2.0 / 1.0 °C	Frost remain less than 1 / 3 of outdoor coil and during melting frost have not abnormal noise.				
	Enclosure Sweat	In 27.0 / 24.3, Out 27.0/24.3 °C	RH80%, operate 4Hr. indoor fan low must be no water drop from air outlet and disposal of condensated water not found.		-	OK	Reference : MSY-KY09VF-TH1 (SAM-24-760-0028) ✓
	Other test (Depend on change point)						
Sound	Overall (SPL)	Cooling mode : In DB27.0 °C, WB19.0 °C Out DB35.0 °C, WB24.0 °C	Indoor side : Fan Rated Hi speed= 43 dBA (930 rpm) not over + 4dBA Fan Super Hi speed= 43 dBA (930 rpm) not over + 4dBA Fan Hi speed= 37 dBA (760 rpm) not over + 4dBA Fan Med speed= 31 dBA (650 rpm) not over + 4dBA Fan Low speed= 25 dBA (530 rpm) not over + 4dBA Fan S.Low speed = 19 dBA (450 rpm) not over + 4dBA Outdoor side : Front side= 52 dBA (960 rpm) not over + 4dBA Back side= 52 dBA (960 rpm) not over + 4dBA		1	OK	Indoor side : Fan Rated Hi speed= 38.3 dBA 932 rpm ✓ Fan Super Hi speed= 38.4 dBA 933 rpm ✓ Fan Hi speed= 32.4 dBA 755 rpm ✓ Fan Med speed= 26.5 dBA 653 rpm ✓ Fan Low speed= 21.0 dBA 531 rpm ✓ Fan S.Low speed = 16.9 dBA 448 rpm ✓ Outdoor side : Front side= 49.5 dBA 964 rpm ✓ Back side= 50.4 dBA 964 rpm ✓
	Abnormal noise	Cooling mode : In DB27.0 °C, WB19.0 °C Out DB35.0 °C, WB24.0 °C	Feeling check no abnormality noise - Pipe length 5 m.		1	OK	- Indoor start cool mode fan low and super low have refrigerant noise and compressor noise level low. ✓ - Indoor stop cool mode have refrigerant noise and compressor noise level low. ✓ - Outdoor start and stop cool mode have compressor noise and refrigerant noise level low. ✓
	Overall (SPL)	Heating mode : In DB20.0 °C, WB- °C Out DB7.0 °C, WB6.0 °C	Indoor side : Fan Rated Hi speed= dBA (rpm) not over + 4dBA Fan Super Hi speed= dBA (rpm) not over + 4dBA Fan Hi speed= dBA (rpm) not over + 4dBA Fan Med speed= dBA (rpm) not over + 4dBA Fan Low speed= dBA (rpm) not over + 4dBA Outdoor side : Front side= dBA (rpm) not over + 4dBA Back side= dBA (rpm) not over + 4dBA				
	Abnormal noise	Heating mode : In DB20.0 °C, WB- °C Out DB7.0 °C, WB6.0 °C	Feeling check no abnormality noise - Pipe length 5 m.				
	Other test (Depend on change point)						
EMC	Impulse noise		Function operate are not abnormal When voltage supply £ 1.8 KV				
	Operating characteristic		Operate check confirm by IB				
Safety	Other test (Depend on change point)						
	Insulation resistance		10 M.Ω or more / 500 V.		1	OK	100 M.Ω ✓
	Dielectric Withstand Voltage		Cut off current 100 mA : Basic Insulation 1000 V / 1 min Supplementary Insulation 1750 V / 1 min Reinforced Insulation 3000 V / 1 min		1	OK	PASS ✓
	Earthing continuity		25A between earthing and metal parts resistance not exceeding 0.1 W		1	OK	PASS ✓
	Leakage Current		Not exceed 2 mA. per Kilowatt of max power input. (2.96 mA)		1	OK	PASS ✓
	Other test (Depend on change point)						
Final Inspection	DT Packing Inspection		Accessories , IB , IM , Guarantee must be packing and check		1	OK	PASS ✓
	Fuction Test at Production Line		Can operate comply with model funtion		1	OK	PASS ✓
	PQI Packing Inspection		Accessories , IB , IM , Guarantee must be packing and check		1	OK	PASS ✓

2. Problem

No.	Detail	Cause	Improvement	Judgement
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COMMENT :

ตารางที่ 7 เกณฑ์ระดับประสิทธิภาพพลังงานเครื่องปรับอากาศ ชนิด Variable Speed/Inverter

ขนาดเครื่องปรับอากาศ	ค่าประสิทธิภาพ : SEER (ดัชนีชี้วัดประสิทธิภาพ)					
	1	★	★★	★★★	★★★★	★★★★★
ไม่น้อยกว่า 8,000 บีทียู/ชั่วโมง (≥27,296 บีทียู/วัน)	17.06 - 18.55	18.56 - 20.05	20.06 - 21.55	21.56 - 23.05	23.06 - 24.55	≥ 24.56
ไม่น้อยกว่า 8,000 - 12,000 บีทียู/ชั่วโมง (≥27,296 - 40,344 บีทียู/วัน)	16.04 - 17.15	17.16 - 18.27	18.28 - 19.38	19.39 - 20.50	20.51 - 21.62	≥ 21.63
ไม่น้อยกว่า 12,000 - 18,000 บีทียู/ชั่วโมง (≥40,344 - 61,616 บีทียู/วัน)	14.00 - 15.49	15.50 - 16.99	17.00 - 18.49	18.50 - 19.99	20.00 - 21.49	≥ 21.50